Experiment No. 3

Title:

Manual Management of Storage Devices in Linux: Identification, Mounting, and Unmounting

Aim:

To understand and demonstrate how to identify storage devices in Linux, manually mount and unmount file systems, and comprehend the importance of mount points and device management in system administration and cybersecurity contexts.

Theory:

Linux File System and Mounting

Linux represents hardware devices as files located in the /dev directory. Common examples include:

- /dev/sda, /dev/sdb – Hard disks
- /dev/sda1, /dev/sdb1 – Disk partitions

Mount Points

A mount point is an empty directory where a device's file system is attached.

Example:

- /mnt /media.

Unmounting Devices

Unmounting safely disconnects the device from the file system, ensuring all data is written properly and preventing corruption.

Commands Used

Commands	Description
lsblk	List block devices
fdisk -l	Display disk partition information

mount	Mount a file system
umount	Unmount a file system
df -h	Display mounted file systems
mkdir	Create mount point
blkid	Display block device attributes

Procedure / Steps

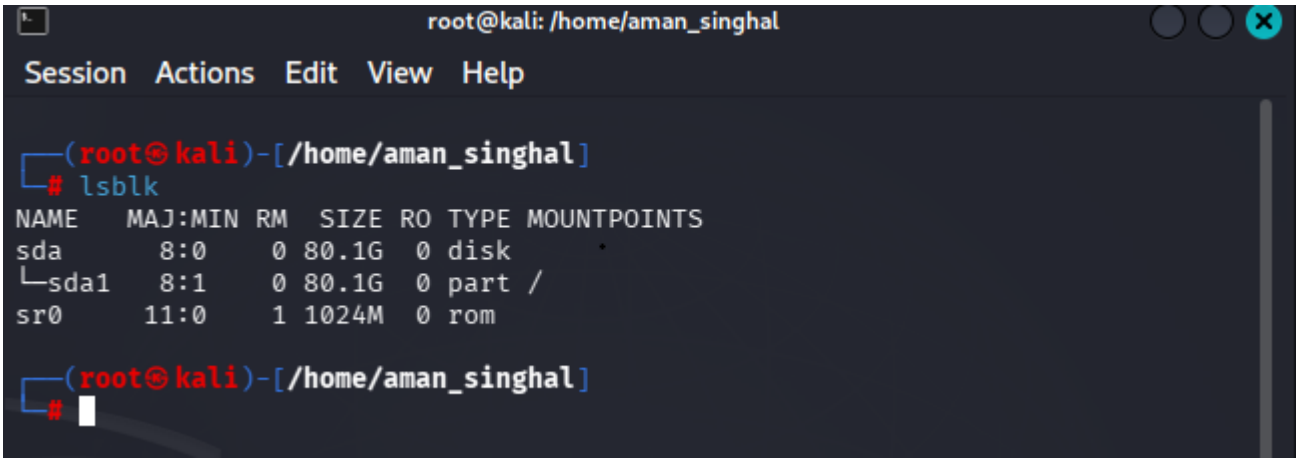
A) Identifying Devices:

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: List available block devices:

- lsblk



Step 3: Display detailed disk information using:

- `sudo fdisk -l`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# fdisk -l
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd094a0b4

Device     Boot Start       End   Sectors  Size Id Type
/dev/sda1  *     2048 167968749 167966702 80.1G 83 Linux

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

B) Creating a Mount Point:

Step 4: Create a directory to be used as a mount point:

- sudo mkdir /mnt/usb

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# mkdir /mnt/usb

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

C) Mounting a Device:

Step 5: Mount a partition (example /dev/sdb1) to the mount point using:

- sudo mount /dev/sdb1 /mnt/usb

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# mount /dev/sda1 /mnt/usb

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 6: Verify the mount Using:

- `df -h`
- `mount`

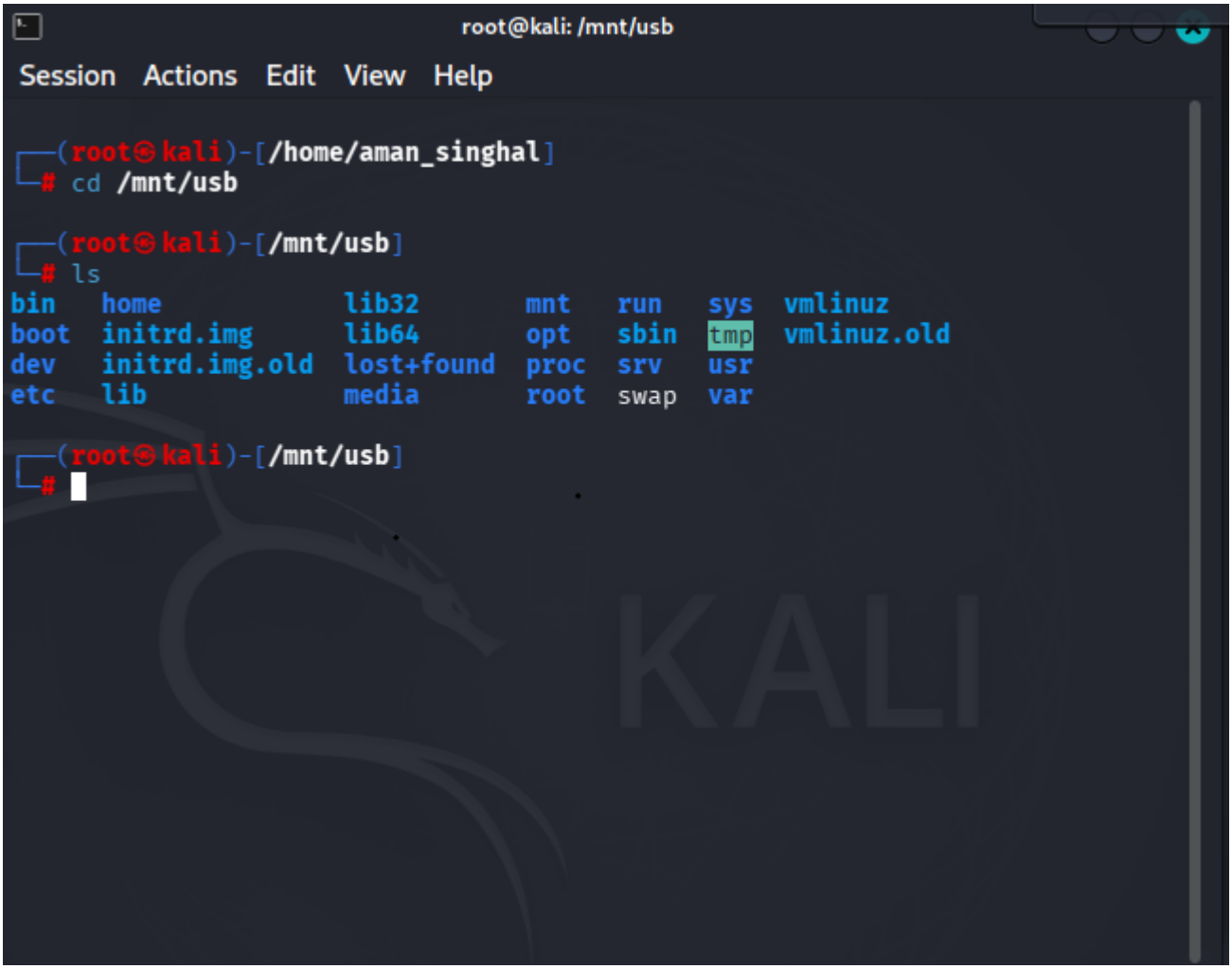
```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            4.8G     0  4.8G   0% /dev
tmpfs           985M  1.2M  984M   1% /run
/dev/sda1        79G   21G   54G  28% /
tmpfs           4.9G  4.0K  4.9G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.servi
ce
tmpfs           4.9G   12K  4.9G   1% /tmp
kali_linux      476G  239G  238G  51% /media/sf_kali_linux
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           985M  116K  985M   1% /run/user/1002

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help
┌──(root㉿kali)-[/home/aman_singhal]
└─# mount
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
udev on /dev type devtmpfs (rw,nosuid,relatime,size=4937480k,nr_inodes=123437
0,mode=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=600,ptmx
mode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,noexec,relatime,size=1008636k,mode=
755,inode64)
/dev/sda1 on / type ext4 (rw,relatime,errors=remount-ro)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,re
latime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdel
egate,memory_recursiveprot,memory_hugetlb_accounting)
none on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=41,pgrp=1,t
imeout=0,minproto=5,maxproto=5,direct,pipe_ino=5019)
none on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev
,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,inode64,noswa
p)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,rela
time)
```

D) Accessing the Mounted Device:

Step 7: Navigate to the mounted directory Using:

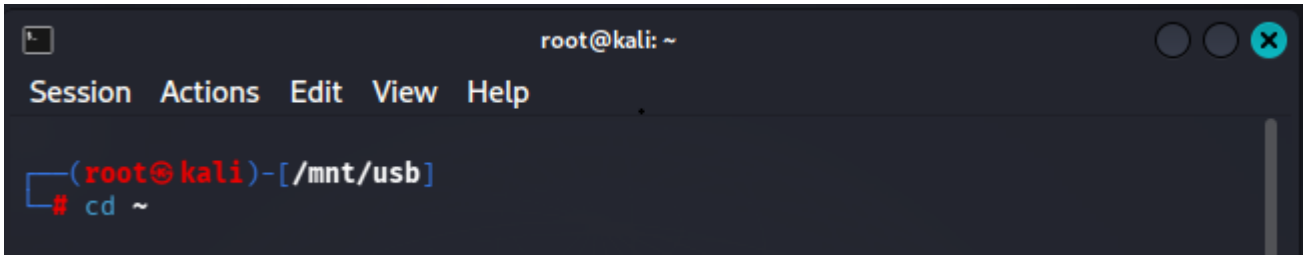
- cd /mnt/usb
- ls



E) Unmounting the Device:

Step 8: Exit the mounted directory Using:

- `Cd ~`



Step 9: Unmount the device using:

- `sudo umount /mnt/usb`

```
┌──(root㉿kali)-[~]
└─# umount /mnt/usb

┌──(root㉿kali)-[~]
└─#
```

Step 10: Verify unmounting Using:

- `df -h`
-

```
root@kali: ~
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[~]
└─# df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            4.8G     0  4.8G   0% /dev
tmpfs           985M  1.2M  984M   1% /run
/dev/sda1        79G   21G   54G  28% /
tmpfs           4.9G  4.0K  4.9G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.servi
ce
tmpfs           4.9G  8.0K  4.9G   1% /tmp
kali_linux      476G  239G  238G  51% /media/sf_kali_linux
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           985M  116K  985M   1% /run/user/1002

┌──(root㉿kali)-[~]
└─#
```

Outcomes:

- Identify storage devices (hard disks, USB drives, external storage) using Linux commands such as lsblk, fdisk -l, and blkid.
- Understand the concept of mounting points and the process of manually mounting file systems using mount and umount commands.
- Safely unmount storage devices to prevent data corruption.
- Understand how Linux integrates storage devices into a unified directory structure.
- Recognize the implications of device management for system administration and cybersecurity, including secure handling of removable media.
- Troubleshoot common issues related to device mounting and access permissions.

Result:

- Successfully identify all connected storage devices using Linux commands.
- Correctly mount and unmount storage devices to specific directories.
- Verify that mounted devices are accessible and data can be read or written.

- Gain practical understanding of how Linux handles storage devices and file system integration.
- Recognize the significance of proper device management in maintaining system security and stability.